

1. Given $a = \{1, 2, 3, 4\}$ and $b = \{3, 4, 5\}$, print their union, intersection, $a - b$ and $b - a$, each in sorted order.

Hint: `sorted()` around every result, since a set has no order of its own.

2. Using the same two sets, print how many distinct values there are across both, and how many values they have in common.

Hint: `len()` on the union and on the intersection.

3. Given $py = \{\text{"asha"}, \text{"ravi"}, \text{"meera"}\}$ and $sql = \{\text{"ravi"}, \text{"kiran"}\}$, print the names of students taking both courses, the names taking Python only, and the total number of distinct students.

4. Given two lists of student names with some overlap, print the names present in exactly one of the two lists, in sorted order.

Hint: convert both lists to sets first, then use the symmetric difference operator $\wedge$.

5. Given $a = \{1, 2, 3, 4\}$, print True or False for each: is $\{1, 2\}$ a subset of a , does a contain every item of $\{3, 5\}$, and do a and $\{9, 10\}$ share nothing at all?

Hint: $\leq$ tests subset, $\geq$ tests superset, and `isdisjoint()` answers the last one.